



Digital game-based learning for transgender identity awareness: a qualitative study in Spanish social education degree

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Abstract

Game-based learning (GBL) has demonstrated its effectiveness in education by addressing challenges such as apathy, motivation, and active student engagement. In this article, the narrative video game “A Normal Lost Phone” has been employed to raise awareness among university students about transgender and gender diverse (TGD) individuals. The project involved 35 undergraduate students in Social Education from a university in Spain (29 females, 5 males, and 1 non-binary individual), with an average age of 20.52 years ($SD=1.82$) during the academic year 2022/2023. The workshop’s main objective was to enhance undergraduate students’ training in social education and better prepare them for their future careers. The results highlight that video games can promote sensitivity and understanding of sexual and gender diversity among university students, which could have significant implications for professional training in this field. In conclusion, GBL presents promising opportunities for transforming educational practices. The integration of cooperative workshops into teaching methodologies can have a positive impact on student engagement, learning outcomes, and attitudes. Further research is needed to comprehend and effectively incorporate GBL into teaching practices fully. By harnessing the educational potential of games, we can create more engaging and effective learning environments that equip students with the necessary skills for their future careers in social education.

Keywords Educational games · Video games · Gender minorities · Community education · Transgender and gender-diverse

Introduction

Transphobia understood as prejudice and discrimination against transgender individuals constitutes a social and cultural phenomenon rooted in a rigid and binary conception of gender (Kearns et al., 2017). This conception posits that

gender is determined by biology and that only two categories exist: male and female. Consequently, anyone who does not conform to these gender norms is deemed abnormal or ill, leading to social stigma that negatively impacts their quality of life and well-being (Borgogna et al., 2019).

Stigma is a multifactorial phenomenon fueled by stereotypes and prejudices towards transgender individuals. Common stereotypes include portraying transgender individuals as dangerous, deviant or mentally ill (Howansky et al., 2021). Stigma severely affects the quality of life and self-esteem of transgender individuals, hindering their social inclusion. Consequently, this negatively affects their employment prospects, economic independence, and ability to form intimate solid friendships, often leading to a cycle of violence. This personal situation can profoundly affect the self-esteem and mental health of transgender individuals, as stigma and discrimination can lead to depression, anxiety, social isolation, and, in extreme cases, suicide (Pellicane & Ciesla, 2022).

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The Gender Minority Stress Model (GMS Model) (Hendricks & Testa, 2012) expounds on the reasons behind the higher prevalence of mental health challenges among transgender and gender diverse (TGD) individuals, building upon the foundation of the Minority Stress Model (Meyer, 2003). The GMS Model illustrates how gender minority groups grapple with additional stress stemming from their marginalized status (Testa et al., 2015). This encompasses both personal stressors, such as self-stigma, and societal stressors, including experiences of discrimination (Scandurra et al., 2020). The GMS Model asserts that the increased occurrence of mental health difficulties among TGD individuals is intricately connected to the stressors arising from their association with a marginalized community (Mezzalana et al., 2023).

One of the professional domains capable of mitigating these adverse effects operates within the sphere of healthcare. Previous studies mention that self-concept can improve when healthcare professionals hold positive attitudes toward the transgender community, leading to more positive and compassionate treatment (Petrey et al., 2022). Reducing structural, interpersonal, and individual stigma experienced by transgender patients requires institutions to provide learning opportunities for future healthcare providers (McCave et al., 2019).

To reduce prejudices among professionals working with vulnerable populations, the importance of socio-educational actions in initial teacher training has been observed (Cao et al., 2020; Martino et al., 2020). These proposals include trans-affirmative policies, raising awareness of the queer community, debates, and sensitization talks. Additionally, some associations and groups bring LGBTQ intervention programs to schools, such as the Safe Schools Coalition, GLSEN, Gay-Straight Alliance, The Trevor Project, or It Gets Better Project (Abreu et al., 2021; Leonardi & Staley, 2015). These educational projects focus on preventing bullying or cyberbullying, as research shows that LGBTQ youth are exposed to more negative experiences in school than their heterosexual and cisgender counterparts.

As mentioned in the previous paragraph, in the fight against stigma traditional talks, training and debates have been used, however other emerging methodologies have also a capacity to favor change in stereotypes and prejudices that sustain the stigma. Among these methodologies are the digital game-based learning (DGBL) methodology. The DGBL allows the adaptation and creation of new pedagogical content, adapting the curricular contents to game mechanics and dynamics, so that students can acquire knowledge and skills through solving challenges and problems presented within the games (Manzano-León et al., 2021). The DGBL is an active methodology that allows the people involved to interact, achieving greater immersion,

motivation and engagement regardless of the subject matter utilizing video games. (Jabbar & Felicia, 2015; Martínez et al., 2022). Additionally, such games have been found to promote the display of altruistic behaviors (Gentile et al., 2009).

A previous systematic review states that traditionally the representation of the transgender and gender diverse (TGD) community in video games was relegating the LGBTQ community to secondary roles, neglecting it (Thach, 2021). This representation often conveys a message in accordance with stereotypical and prejudiced social norms, without taking into account the uniqueness of the individual, being portrayed as dangerous, monstrous and evil entities (Halberstam, 2018). However, progress has been made in recent years, with prominent titles featuring LGBTQ characters as protagonists of their narratives. These games offer positive and organic perspectives on gender diversity. Notable examples include “*Tell Me Why*”, “*The Last of Us 2*” and “*Celeste*”. However, further research remains imperative to explore the potential of video games to promote awareness of TGD and LGBTQ in educational settings (Rosiek et al., 2017). In this context, the use of video games and serious games within education proves valuable for multiple objectives. It raises awareness among young individuals, reduces stigma, and furthers their societal inclusion and well-being (Rodríguez-Ferrer et al., 2022).

Purpose of the research

Recent research has shown the effectiveness of gaming strategies in raising awareness of different social causes, such as stigma towards people with severe mental disorders, gender equality, sexual violence, mental health or bullying (Moukram et al., 2023; Rodríguez-Ferrer et al., 2023), relating that this can occur due to the immersion produced by the narrative and game mechanics (Aura et al., 2021). However, more research is needed to understand its potential benefits for teaching and raising awareness of LGBTQ people, particularly TGD people, since the capacity of video games for the exploration and validation of gender identity as well as immersion and escapism for TGD adolescents has been observed (McKenna et al., 2022). The present study proposes an investigation of the use of DGBL in higher education with undergraduate social educators, as it is a profession whose professional opportunities may be related to improving the quality of life of people at risk of LGBTQ social exclusion. In this context, the originality of the present study is to explore the application of a narrative video game to raise awareness of TGD people in higher education.

The research questions have been the following:

- RQ1) Can the narrative of a LGBTQ friendly video game raise awareness of TGD people in higher education?
- RQ2) What is the learning assessment of a DGBL methodology for university students?
- RQ3) Can Social Education students acquire professional competencies through video games in the classroom?
- RQ4) What disadvantages do university students perceive about video games and their implementation in the classroom?

Method

Participants

A convenience sample was chosen to select the participants. The inclusion criterion for the experimental group was the willingness to participate in the training workshop organised by one of the leading researchers.

The dataset for the analysis consisted of 35 (83.3%) undergraduate students majoring in Social Education from a university in Spain. The sample comprised 29 women, 5 men, and 1 non-binary individual, with an average age of 20.52 years ($SD=1.82$) during the academic year 2022/2023. A majority of women is observed in the sample because the education sector in Spain is dominated by women (González Alba et al., 2021).

These students were enrolled in a subject titled Social Skills in Social Education, which was part of their second-year curriculum. This subject was a mandatory component of the program and formed an integral part of the compulsory curriculum. However, this specific activity within the course remained optional. Out of a total of 42 students enrolled in the course, 35 students opted to participate in this specific activity. All students who participated in the activity, engaged in the investigation and willingly took part in the focus groups.

Before the study commenced, all participants were provided with comprehensive information about the project. They provided written informed consent following the guidelines outlined in the Declaration of Helsinki (Carlson et al., 2004). Prior to data collection, students were thoroughly briefed about the study's nature, and their anonymity was assured. The study received approval from the Teaching Innovation Group "EDU-INNO" and was authorized by the University of Vigo.

Instruments

Focus groups have been used, designed to collect perceptions about the area of interest in a non-threatening environment. Recently, focus groups have been used in various health related applications due to their simplicity and convenience (Kreuger & Casey, 2000).

Five focus groups were carried out, in which the students were randomly assigned to each of them. The average duration for each of these groups was 40 min, which allowed the theoretical saturation of the categories necessary for the analysis. For the data collection, a participant observation was used in which the data were collected by two external observers in a systematic way. To ensure participant anonymity, a unique code was assigned to each individual.

Procedure

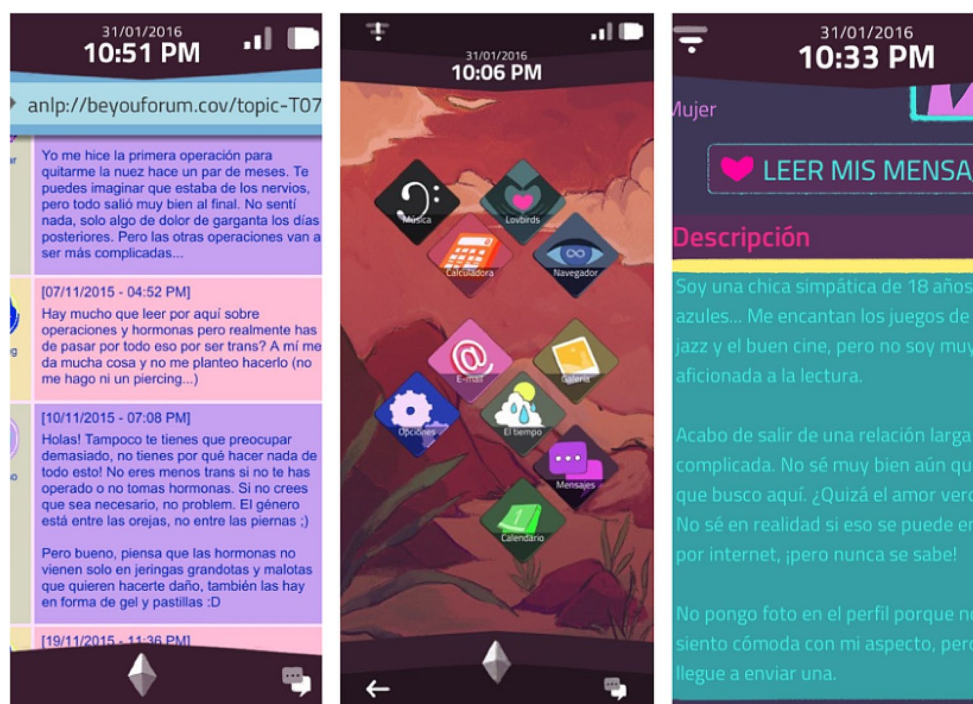
The DGBL methodology was applied in the regular timetable of the practical part of the course during two sessions of two hours each. The activity consisted of playing a video game cooperatively in teams of 4 to 6 students, and carrying out a critical essay on it after ending the game. After the activity, they were offered to participate voluntarily in focus groups to deepen their opinions about the experience. It was made clear to students that their participation in the focus groups did not impact the subject grade.

Previously, a narrative review was conducted on video games with transgender characters, highlighting a systematic review study that analyzed 63 games released between 1988 and 2019 featuring at least one transgender character. This review revealed four stereotypical trends in how video games portray transgender characters, namely, dysphoria/physical transition, mentally ill killers, trans-shock/revelation, and ambiguity (Thach, 2021).

The video game "*A Normal Lost Phone*" was chosen because of its trans-friendly perspective, its relation to the subject matter under study, and its low price, facilitating its acquisition and implementation in the classroom (See Fig. 1).

A Normal Lost Phone is a puzzle video game in which the gameplay is presented as a mobile phone interface. The player has to search through it and find out what happened to its owner. For that, you will have to inquire into your messages, your email or your latest searches on the internet. The game was created in 2017 and developed by Accidental Queens. A sequel titled *Another Lost Phone: Laura's Story* was published in the same year. *Normal Lost Phone* presents the story of Sam, a teenage boy living in Melren. As the game progresses, players will discover several vital secrets, such as a friend of Sam's attempt to rape a friend or the protagonist's birthday. Players also find that Sam is a trans

Fig. 1 Screenshots of A normal lost phone. *Note:* Screenshots from “A Normal Lost Phone” by Accidental Queens, 2017. Reproduced with permission of the authors



bisexual woman named Samira who has been hiding this from those closest to her. Players will eventually find Sam’s dating profiles, one where he describes himself as male and one female, and other LGBTI support forums where Sam accepts who he is.

To make matters worse, he discovers that his parents and girlfriend Melissa are very intolerant of the LGBT community, leading Sam to break up with Melissa. Sam finally plucks up the courage to talk to his friend Alice, who accepts his gender identity. As a last resort, after his father gives him a motorcycle, he throws away his mobile phone and leaves home to live in a new city. The only person he tells the plan to is Alice, who congratulates Samira on taking control of her life and tells her that anyone who finds her mobile phone will erase all the information. According to Alice’s message, the game ends when the player erases the phone data.

Data analysis

Data analysis was done through a thematic analysis (Braun & Clarke, 2013). A detailed analysis of the categories was carried out with labels.

The threshold for including a sub-theme was defined using a systematic approach, considering both the frequency of occurrence and its relevance to the research objectives. Specifically, the threshold was set at a minimum of three instances within the set of focus groups, ensuring that only sub-themes meeting this criterion were included in the analysis.

Related codes were grouped into sub-themes and overlapping sub-themes were grouped. This analysis was carried out independently by two researchers. After the individual analysis, they pooled the codes and, in case of disagreement, a third researcher intervened.

The extracts of focus groups have been translated and are reproduced here in English. The information was transcribed and incorporated into ATLAS.ti software (version 9, ATLAS.ti Scientific Software Development GmbH, Berlin, Alemania).

Results

The data obtained from the focus groups examined in the study were conceptualised under 3 themes (11 sub-themes): Narrative and gameplay elements within DGBL, Training through the video game, and Raising awareness towards the TGD people (Tables 1, 2 and 3).

Coinciding with previous research regarding the motivational capacity that games and gamification provide to university students (Macías-Guillén et al., 2021), 97.3% of the students in the focus group showed high interest in learning and were more willing to participate in activities related to the video game actively. They enjoyed the experience of playing the video game and were more willing to engage in the activity actively. Similarly, the students perceived that the video game created a fun, creative, and engaging learning environment.

Table 1 Narrative and gameplay elements within DGBL theme

Main theme	Sub-themes	%	Quote example
Narrative and gameplay elements within DGBL	Motivation	97,3%	“I liked it a lot, it is a way to learn and sensitize you on the subject to deal with LGBTI +, since you enter the game and it is as if you lived it in the first person, everything that is constructive is positive, since it helps you to inform yourself about the subject and sensitize you and clarify your opinions on this topic. It opens your eyes to what reality is” (FG1S3) “For me, gamification is a positive experience in the classroom. It is a new and different way of working that gives us, the students, a new motivation to learn. In addition to a new approach to learning where, at least personally, I learn better and faster, as I memorize procedures and patterns that can serve me in the future” (FG2S2) “[DGBL] captures our attention and interest more than a traditional class” (FG5S6)
	Fun	16,2%	“In my opinion, with the realization of this activity I learned other ways of teaching since no one had ever used video games to teach me anything before. In addition, the tension and autonomy that made us take the game made it more interesting and fun” (FG4S9)
	Creativity	54,1%	“I think it was very positive [the DBGL] because it supposes the exit of the routine and perhaps the motivation to come to the class, to creativity and to the collaborative work, which in many cases causes the feeling of belonging to a group and of learning without being studying and really causes you to want to go to the class” (FG3S5)
	Learning	70,3%	“I have learned that activities related to social education are not merely theoretical or practical, at the level of working with people, but that something as current as video games can help us intervene in a collective with LGBTI + and especially in TGD people” (FG2S1)
	Previous prejudices about video games	27,0%	“To the games that I played at the time none [has no use], since of those (5 years ago) FIFA was only the male representation, today I do not know if they already put the women’s league, yes here there is no equality, values 0, in the GTA and better not to mention the type of game like Call of Dutti [Call of Duty], gun games only,” (FG1S8) “Video game addiction is something that is there and exists without a doubt” (FG4S7)
	Possible improvements in the application of the methodology	45,9%	“I personally enjoyed the activity carried out in the class, and I believe that more classes of this type could be held, even if time is scarce, since I think it is very necessary for the motivation and learning of the students” (FG3S6) “I really liked that gamification was used, in this case through a video game since I had never tried it before, and it turned out to be a very entertaining video game. The only one that could change would be to apply gamification in more subjects and in a somewhat more habitual way” (FG5S3)

Note: FG = Focus group (Number), S = Student (Number)

Lastly, 45.9% of the students suggested an increase in instructional hours for this specific activity and an increase in playful methodologies in the classroom. This suggests widespread acceptance of these methodologies among university students.

As observed in the frequencies of Table 2, students have perceived that video games have been beneficial to their academic development, both in terms of the specific content about TGD people and in the specific training related to their degree. To a lesser extent, video games have also been seen to contribute to their social skills and cooperative abilities.

Finally, it is worth highlighting that using the video game *A Normal Lost Phone*, which features a transgender protagonist, has fostered a greater awareness and sensitivity towards this community among students. The students emphasize that the game allows you to “experience that reality,” to understand and internalize attitudes and prejudices towards this community.

Discussion

This study presents the results of a qualitative investigation on the potential of video games as pedagogical tools for undergraduate social educators. Specifically, the research explores students’ perceptions regarding using “*A Normal Lost Phone*” to promote awareness and learning about the stigma towards TGD individuals. The research on the application of the video games to raise awareness about the TGD community among social education undergraduate students can have several important implications.

Firstly, the results show that university students perceived the video game has fun mechanics and an attractive narrative, allowing them to empathise with the character and understand the TGD community. This perceived fun can be an essential factor for immersion in the activity and, consequently, for learning opportunities, as suggested by Tisza and Markopoulos (2023).

Secondly, this exploratory study suggests that video games may promote sensitivity and understanding of sexual and gender diversity among university students, which could have significant implications for the training

Table 2 Training through the video game

Main theme	Sub-themes	%	Quote example
Training through the video game	Specific training on professional treatment towards the TGD collective	73,0%	“You can prevent these problems [prejudices and negative attitudes] by showing the social reality experienced by LGTBI+ people” (FG2) “Through a video game you can transmit much more, transporting a person to a certain reality as if it had been their own or as something very close, so it is very helpful when transmitting experiences and values with the aim of educating effectively in this subject” (FG3)
	Cooperation and development of social skills	29,7%	“Video games can help us to establish communication with the participants, to be able to dialogue with them, either through chat or audio” (FG1) “Playing video games helps to dialogue and communicate with other people, to tolerate frustration, increases creativity and strengthens relationships with people you don’t know” (FG4) “The management of our emotions since when learning through video games we have to learn to self-manage the emotions that make us feel” (FG6)
	Social Education training	91,9%	“Just as there are prevention programs, informative talks, you can create games to make those programs or talks more enjoyable, because in the end that a person is talking to you for 1 hour ends up boring you, instead almost everyone likes to play, so instead of a person giving you the talk orally, Create a game in which it is indirectly telling you the same thing as that talk or program but in a playful way and so the people who participate can get more things clean than the title of the program.” (FG1) “Social education can promote social transformation and positive change in society through actions on the different affected groups and can be helped by gamification since in the actions can be made dynamics or gamified activities that help to achieve the desired objective” (FG2) “In the first place, the main one in this context would be to provide quality and depth to social educators, so that they know how to convey a message. In an indirect way, because I believe that it is really how the transformation can take place, with the change of thought. Having clarified this, I believe that gamification, well designed, can educate and transform, especially at an early age, but it must be addressed in the right way. I think that, in conclusion, what is missing is to stop and look for real quality, because video games are a useful tool. Why can’t you popularise a game that really engages in the sense that it’s really entertaining and that works on social education?” (FG3)

Note: FC = Focus group (Number), S = Student (Number)

Table 3 Raising awareness towards the TGD collective

Main theme	Sub-themes	%	Quote example
Raising awareness towards the TGD collective	Social reality of TGD people	94,6%	“I found it very interesting to put yourself in his situation and see how I was Sam, and you only get that if you are involved in the game, and you see how the family behaved with her and her friends, you do not know what you feel and you seek help to discover who you really are” (FG5) “This activity reflects a reality shown through a video game, which shows situations that TGD people can usually experience” (FG2) “Personally, I can say that I learned a little more about the context that a TGD person may be going through, both the family background and structures that cause the cover-up of their situation and the companies and interactions as milestones that can transform the lives of these people who feel in the process of change or not satisfied with their current life and identity. It helped me a lot in that sense since, despite vaguely knowing the technicalities regarding this group, I was unaware of the reality prior to a change and the gigantic difficulties that a person can face” (FG3)
	Violence prevention	59,5%	“If it serves to educate and raise awareness of gender-based harassment because, as I said previously [previous comment: LGTBI+ themed video games can serve to educate, sensitize and raise awareness of the situations that TGD people live in their day to day, showing it through conversations and videos or forums that these people consult, In the case of this video game], shows the conversations she had with her environment, tells the situation she lived with her family, how she hid her friends and the places where she went, as well as the fear of being “discovered” by her parents and how she hid everything through a forum and a secret app where she could be herself” (FG6) “It is very important to sensitize students about gender diversity and the prevention of violence against the TGD community” (FG2)

Note: FC = Focus group (Number), S = Student (Number)

of professionals in this field. It is highlighted that the ability to simulate other characters’ experiences in video games can facilitate a deeper understanding and empathy towards their realities. Despite the growing number of studies on

non-heterosexual characters in video games (Lacey, 2023), the pedagogical use of such games has not been extensively explored. This study sheds light on the potential of video games as tools for promoting awareness and sensitivity

towards TGD individuals, which is crucial for students in the social and health fields as future professionals. Video games could be an attractive and effective tool for fostering learning and reflection on complex issues such as gender identity and sexual diversity. This could open up new avenues for innovation in education and other fields yielding favorable results (Yılmaz & Griffiths, 2023).

Additionally, the research on applying the video game “A Normal Lost Phone” could have significant educational implications. One notable finding of this study is that students perceived that they learned significantly when playing the game in the classroom. Playful strategies such as gamification and the use of video games aim to promote a sense of engagement with learning among students, which consequently leads to enhanced learning outcomes (Macías-Guilén et al., 2021). The results of the present study align with this argument, as evident from the obtained categories. University students perceive this workshop as valuable for their professional development, providing an alternative perspective for addressing classroom topics, such as bullying and cyberbullying, and practical knowledge of ICT (Information and Communication Technology) resources. Therefore, it has become even more critical for university teachers to address the changing times and employ emerging technologies and participatory practices to create motivating learning environments for students. Furthermore, participants note to a minor degree that the cooperative approach of DGBL has provided them with the opportunity to engage with their peers and enhance their social skills. This finding aligns with previous results (Granic et al., 2014; Nino & Evans, 2015).

Likewise, raising awareness about TGD issues can have significant implications in the fight for the rights of TGD individuals. If university students can better understand the experiences and needs of these individuals, they may be more likely to support policies and actions that promote equality and social justice (Yi et al., 2023). The research on the application of the video game “A Normal Lost Phone” could also have implications for the industry. Using video games to promote awareness and understanding of sexual and gender diversity can potentially encourage the development of video games that address other important social issues.

Limitations and further research

Although the study results demonstrate promising developments in increasing awareness of TGD among socio-educational professionals, several limitations exist. Firstly, the sample size is relatively small, which restricts the generalization of the results. In future research, larger samples

could be used, and comparisons could be made between bachelor’s and master’s degree students. In addition, the study participants were all university students, so the results may not apply to other socio-educational professionals or the general population. Another limitation is the method of sample selection. Due to the data being collected in the natural context of the classroom, gender and age analyses were not possible, as the groups mainly consisted of women between 20 and 25 years old. This limits the study’s ability to assess whether the program had a different effect on men or individuals from different age groups. Thirdly, another potential limitation is the voluntary nature of the workshop. It was designed this way to mitigate resistance or reluctance that participants might have, aiming for more honest and sincere responses and engagement. However, this voluntariness could constrain representativeness and lead to selection bias. Future quasi-experimental studies could explore the impact on the studied variables by distinguishing between experimental groups involving voluntary activities and those involving mandatory activities. Additionally, the study did not collect data on participants’ previous knowledge or experiences with TGD issues, which may have influenced their response to the intervention.

Furthermore, the program was applied over four hours, so the long-term effects of this methodology could not be determined. The program’s effects may diminish over time, or other factors may influence its impact in the longer term. Future research could focus on longitudinal studies of video games as a tool for raising awareness among university students and other populations, to determine the long-term effectiveness of this methodology. Lastly, the study focused specifically on the use of video games as a tool for raising awareness of TGD gender-diverse issues. While this approach is innovative, other potential interventions can be studied with video games, such as workshops, training programs, or educational campaigns. Future research could compare the effectiveness of different interventions and explore the potential benefits of combining different strategies to raise awareness.

Conclusions

This study’s findings demonstrate social education students’ attitudes towards using a video game in the classroom to promote awareness and learning about TGD diversity. Students generally showed enthusiasm towards the game, identifying its enjoyable, team-oriented, and creative features and its capacity for enhancing awareness and learning for their professional development.

These results suggest that video games have the potential to be an effective tool in promoting sensitivity and

understanding towards the TGD community among university students. The interactive nature of video games and their ability to provide immersive and engaging experiences may make them particularly effective in fostering empathy and understanding towards TGD individuals. Continuing research in this field is crucial for assessing the efficacy of these approaches and uncovering fresh methods to advance equality and social justice for everyone, including individuals within the transgender and gender diverse community.

Data availability The data supporting this study's findings are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declarations

Ethics approval and consent to participate All participants received information about the project and gave written informed consent following the Declaration of Helsinki. Before data collection, students were informed of the nature of the study and guaranteed anonymity. The Teaching Innovation Group “EDU-INNO” authorised the study, approved by the University of Vigo. The article's authors declare that they have no conflict of interest regarding the publication and research carried out.

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